

KalariSena: Learning Physics-Grounded and Recoverable Kalaripayattu Skills for Humanoid Robots

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Abstract

Humanoid robots can increasingly mimic diverse human motions, yet **highly dynamic skills remain difficult to transfer faithfully across embodiments**. This gap is acute for movements involving deep stances, rapid weight transfer, asymmetric support, abrupt contact changes, and large whole-body momentum: a trajectory may be kinematically plausible while remaining physically unstable, and a state that is feasible now may already be incompatible with its intended continuation. We study this problem through **Kalaripayattu**, an ancient Indian martial art built around structured, highly dynamic whole-body movement, and introduce **KalariSena**, a framework for physics-grounded and recoverable humanoid skill execution.

KalariSena combines three components. (i) **Structured skill representation** encodes movement identity, phase, contact mode, and dynamic state. (ii) **Physics-grounded embodiment projection** maps human motion to the humanoid while enforcing contact consistency, support feasibility, friction, joint/torque limits, and whole-body dynamics. (iii) **Successor-Conditioned Viability** estimates whether the current state can safely complete the ongoing skill and reach its designated continuation. A viability-gated residual controller applies **minimal, intent-preserving corrections** as viability deteriorates, before conservative fallback becomes necessary.

We evaluate KalariSena on **100 hours of live-recorded Kalaripayattu** in simulation and on humanoid hardware, assessing fidelity, physical feasibility, viability, robustness, held-out transitions, and intent preservation. KalariSena improves **skill completion from 66.1% to 90.8%**, reduces **physical constraint violations by 69.0%**, and raises **intent preservation by 36.9 points**, demonstrating robust control that preserves a viable path to the intended future.

KalariSena - at-a-glance

- We formulate *dynamic Kalaripayattu transfer* as a **prospective embodiment problem**: motion may be feasible now yet physically unable to reach its intended continuation under contact, actuation, and momentum constraints. (cf. [Sec. 1](#)).
- We introduce **KalariSena**: *structured skill state + physics-grounded embodiment projection*, preserving movement intent while enforcing contact, support, friction, joint/torque, and whole-body dynamics; physical violations fall from **18.4% to 5.7%**. (cf. [Sec. 3](#) and 4).
- We propose *Successor-Conditioned Viability (SCVC)* with viability-gated residual control, achieving a **214 ms anticipation lead time** for loss of the intended continuation and enabling **minimal, intent-preserving corrections** before fallback. (cf. [Sec. 3](#) and 4).
- On **100 hours** of live-recorded Kalaripayattu, KalariSena achieves **81.6% IPR**, **83.7% held-out transition success**, and **77.6% hardware IPR**, validating prospective control across simulation, perturbations, **held-out transitions**, and physical humanoid deployment. (cf. [Sec. 4](#)).

1 Kalaripayattu: An Ancient Martial Tradition and a Challenge for Humanoid Control

Kalaripayattu as structured whole-body movement. Kalaripayattu is an ancient Indian martial tradition from Kerala built around tightly coupled posture, footwork, balance, transitions, and whole-body coordination ([Zarrilli, 1989](#); [Constantini, 2023](#)). Its movement vocabulary includes *vadivu* (posture), *chuvaduvu* (foot placement and stepping), and *meypayattu* (coordinated body sequences). These are not isolated poses: movement identity depends on **how the body enters a configuration, transfers weight, changes support, redirects momentum, and exits toward the next phase**. Kalaripayattu therefore provides a natural setting for studying structured humanoid control rather than pose imitation alone.

Why is Kalaripayattu difficult for a humanoid? The first challenge is the **human-humanoid em-**

Table 1: **Positioning KalariSena against contemporary humanoid whole-body control.** Prior work individually addresses highly dynamic motion, physics-grounded adaptation, skill transitions, and robustness/recovery. KalariSena jointly targets these capabilities through *prospective skill viability*: predicting whether the intended Kalaripayattu continuation remains achievable from the current dynamic state.

Method	High-dynamic	Physics-grounded	Skill transition	Robust / recovery	Future viability	Core distinction from KalariSena
KUNGFUBOT (Xie et al., 2025)	✓	✓	✗	✗	✗	Highly dynamic martial-art tracking, but without explicit successor-conditioned continuation reasoning.
KUNGFUBOT2 (Han et al., 2025)	✓	✗	✓	✓	✗	Versatile and long-horizon dynamic tracking; does not explicitly predict whether a designated future skill remains reachable.
CoRE (Jeong et al., 2025)	✗	✓	✗	✗	✗	Contact-aware physical refinement improves robot-ready reference motion; no online reasoning about the intended continuation.
NMR (Zhao et al., 2026)	✓	✓	✗	✗	✗	Dynamics-aware human-to-humanoid retargeting, including martial motions; physics correction remains upstream of control.
SONIC (Luo et al., 2026)	✓	✗	✗	✓	✗	Large-scale generalist whole-body motion tracking rather than skill-structured, successor-specific control.
BEYOND-MIMIC (Liao et al., 2026)	✓	✗	✓	✗	✗	Composes agile motion behaviors, but does not explicitly estimate state-dependent viability of an intended successor.
SWITCH (Lau et al., 2026)	✓	✗	✓	✓	✗	Skill graph and online transition search; graph reachability is distinct from probabilistic viability of a specified continuation.
ANY2TRACK (Zhang et al., 2026)	✓	✗	✗	✓	✗	Maintains tracking under disturbances and dynamics shifts, but does not explicitly model preservation of the intended skill future.
KUNGFUATHLETEBOT (Lei et al., 2026)	✓	✗	✗	✓	✗	Highly dynamic martial tracking with autonomous fall recovery; KalariSena targets intervention <i>before</i> skill abandonment.
FIRM / STABLEMIMIC (Xu et al., 2026; Wu et al., 2026)	✗	✗	✗	✓	✗	Fall prevention and structured recovery restore safe/trackable states; our target is pre-failure, intent-preserving continuation.
SAFEFLOW (Cho et al., 2026)	✗	✓	✗	✓	✗	Detects unreliable motion and invokes safe fallback; KalariSena attempts to preserve the ongoing skill while continuation remains viable.
KalariSena	✓	✓	✓	✓	✓	Phase-, contact-, and successor-conditioned control that prospectively predicts and preserves a physically viable path into the intended Kalaripayattu continuation.

✓ = explicitly modeled or evaluated as a primary capability; ✗ = not an explicit objective of the method. “Future viability” denotes prospective, successor-conditioned prediction of whether the intended skill continuation remains achievable from the current robot state.

bodiment gap. Given a human demonstration $X_{1:T}^H$, standard retargeting seeks

$$\mathbf{q}_{1:T}^{\text{kin}} = \arg \min_{\mathbf{q}_{1:T}} \sum_{t=1}^T \mathcal{D}(FK(\mathbf{q}_t), X_t^H),$$

but geometric similarity does not guarantee physical executability. Human and humanoid bodies differ in morphology, joint limits, foot geometry, mass distribution, inertia, and actuation; executable motion must additionally satisfy

$$\mathbf{M}(\mathbf{q})\ddot{\mathbf{q}} + \mathbf{h}(\mathbf{q}, \dot{\mathbf{q}}) = \mathbf{S}^\top \boldsymbol{\tau} + \mathbf{J}_c^\top \boldsymbol{\lambda},$$

with $\mathbf{q} \in \mathcal{Q}$, $\dot{\mathbf{q}} \in \dot{\mathcal{Q}}$, $\boldsymbol{\tau} \in \mathcal{T}$, and $\boldsymbol{\lambda} \in \mathcal{F}(\mu)$.

Kalaripayattu amplifies this gap through four recurring regimes: (i) **deep and asymmetric stances**, which compress joint and torque margins; (ii) **rapid support changes**, including double-to single-support transitions; (iii) **momentum-intensive movements**, such as lunges, pivots, jumps, and sweeping motions; and (iv) **temporally coupled transitions**, where the state produced by one phase determines whether the next remains executable. A trajectory may therefore be visually faithful yet exhibit foot slip, incorrect contact, actuator saturation, or an unstable continuation.

What has contemporary humanoid research already solved? Recent work covers much of this

landscape, as summarized in Table 1. **Dynamic tracking:** KUNGFUBOT (Xie et al., 2025) demonstrates physics-based highly dynamic martial-art tracking, while KUNGFUBOT2 (Han et al., 2025) extends this regime to versatile, long-horizon, and unseen motions. SONIC (Luo et al., 2026) scales general whole-body tracking, and BEYOND-MIMIC (Liao et al., 2026) enables composition of agile motion behaviors. **Physics-aware transfer:** CoRE (Jeong et al., 2025) introduces contact-aware refinement, NMR (Zhao et al., 2026) learns dynamics-aware retargeting, and UMR (Cao et al., 2026) learns cross-embodiment correspondence. **Transitions and robustness:** SWITCH (Lau et al., 2026) uses a skill graph and online search for cross-skill transitions, while ANY2TRACK (Zhang et al., 2026) adapts tracking to disturbances and dynamics shifts. **Recovery and safety:** FIRM (Xu et al., 2026), KUNGFUATHLETEBOT (Lei et al., 2026), and STABLEMIMIC (Wu et al., 2026) address fall prevention or recovery, whereas SAFEFLOW (Cho et al., 2026) detects unreliable motion and invokes conservative fallback.

What remains unresolved? Table 1 makes the remaining gap explicit. Highly dynamic tracking

asks whether a difficult reference can be followed; physics-aware retargeting asks whether that reference is executable; skill-transition methods ask how one behavior can connect to another; and recovery methods ask whether the robot can survive or return to a controllable state. **None of these objectives alone asks whether the specific intended continuation remains physically reachable from the robot’s current dynamic state.** This distinction matters because mechanically successful recovery may still destroy the movement being performed.

Feasibility is not future skill viability. KalariSena therefore distinguishes instantaneous feasibility from prospective continuation:

feasible at $t \not\Rightarrow$ viable for the intended future.

A robot may satisfy current pose, contact, and balance constraints while carrying a velocity, momentum, or support state from which the next phase is no longer reachable. During a lunge-to-one-leg transition, for example, a robust controller may remain upright by inserting an additional step, a skill-switching method may redirect toward another reachable motion, and a recovery controller may return to standing. All can succeed physically while *losing the intended Kalaripayattu continuation*.

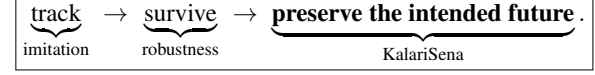
Successor-conditioned skill viability. We formalize this distinction through successor-conditioned skill viability. Given robot state $\mathbf{s}_t$, current skill z_t , phase ϕ_t , contact state c_t , and intended continuation z^+ ,

$$\mathcal{V}^\pi(\mathbf{s}_t, z_t, \phi_t, c_t, z^+) = P_\pi(Y_{\text{safe}} = 1, Y_{\text{complete}} = 1, Y_{\text{succ}} = 1 \mid \mathbf{s}_t, z_t, \phi_t, c_t, z^+).$$

Here, Y_{safe} requires physically valid execution, Y_{complete} requires completion of the current skill, and Y_{succ} requires reaching a valid state for its designated continuation. We use *viability* in an operational, policy-conditioned sense rather than as a formal control-theoretic viability kernel.

From recovery to intent-preserving correction. The distinction above changes the role of recovery. When predicted viability is high, the robot should preserve the demonstrated motion. As viability deteriorates, KalariSena minimally adjusts

support placement, torso orientation, timing, limb velocity, or momentum distribution to restore dynamic margin while retaining the intended skill. Only when continuation is no longer plausibly recoverable does control transfer to a conservative fallback:



Why Kalaripayattu exposes this problem. Kalaripayattu repeatedly traverses dynamically narrow regimes—deep configurations, asymmetric and single support, rapid contact changes, and momentum-intensive transitions—where an arbitrary stabilizing action can alter the movement itself. It therefore exposes a stronger control question than whether a humanoid can imitate a martial-art trajectory or recover after failure: **can it recognize when the intended future is becoming unreachable and correct the current motion while that future is still physically recoverable?** This is the problem addressed by KalariSena.

2 Kalaripayattu and the KalariSena Motion Corpus

Kalaripayattu. Kalaripayattu is an ancient Indian martial tradition from Kerala built around tightly coupled posture, footwork, balance, and whole-body transitions (Zarrilli, 1989; Constantini, 2023). Its vocabulary includes *Vadivu* (stances/postures), *Chuvadu* (footwork), and *Meyipayattu* (coordinated body sequences). Unlike pose-centric motion, a Kalaripayattu skill is defined by **how the body enters, transfers weight, changes support, redirects momentum, and exits toward the next configuration**. This makes it a natural stress test for humanoid embodiment.

100 hours of live-recorded motion. We construct the **KalariSena Motion Corpus** from approximately **100 hours of dedicated live Kalaripayattu recordings**. The recordings preserve complete executions—entry, transition, peak dynamics, and recovery—rather than isolated poses. This is critical because the hardest states frequently occur *between* canonical configurations. As summarized in Table 2, the corpus spans deep stances, directional footwork, asymmetric weight transfer, single-support balance, pivots, explosive advances, aerial transitions, and long-horizon body sequences, exposing complementary support, contact, and momentum regimes.

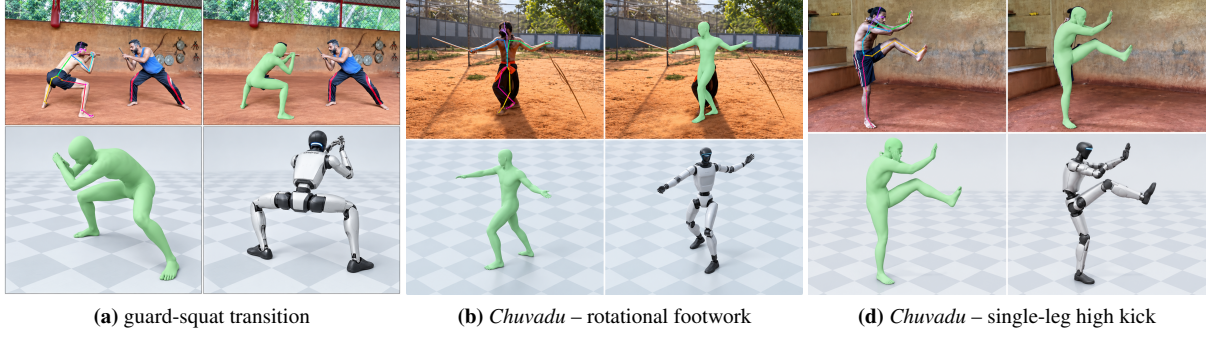


Figure 1: **Live Kalaripayattu to raw humanoid retargeting.** Each example shows *live video + body reconstruction* $\rightarrow$ *normalized 3D motion* $\rightarrow$ *raw Unitree G1 retarget*. (a) guard-squat transition; (b) Chuvadu: rotational footwork; (c) deep crouched stance; (d) Chuvadu: single-leg high kick. The examples span support, rotation, weight transfer, and single-leg dynamics; raw retargeting preserves kinematics but does not guarantee executability.

From video to structured 3D motion. Figure 1 illustrates our data pipeline across representative skills. Given a recorded sequence $I_{1:T}$, we recover temporally coherent 3D human motion

$$X_{1:T}^H = \mathcal{H}_\omega(I_{1:T}) = \{\mathbf{p}_t^H, \mathbf{R}_t^H, \mathbf{r}_t^H\}_{t=1}^T,$$

where $\mathbf{p}_t^H$, $\mathbf{R}_t^H$, and $\mathbf{r}_t^H$ denote 3D joint positions, local orientations, and global root state. Temporal reconstruction is essential for preserving fast kicks, pivots, lunges, and low-stance transitions. We align the recovered motion to the ground, separate root displacement from local articulation, normalize body scale, and retain movement identity, normalized phase ϕ_t , and foot-contact state $\mathbf{c}_t$. Figure 1 shows this progression from live video and pose structure to recovered human geometry and normalized 3D motion.

Human-to-humanoid retargeting. Direct joint-angle transfer is impossible because human and humanoid embodiments differ in skeleton topology, degrees of freedom, limb proportions, and joint ranges. We establish semantic correspondences across the pelvis, torso, shoulders, arms, hands, hips, knees, ankles, and feet, and solve

$$Q_{1:T}^0 = \arg \min_Q \sum_t [\lambda_p \mathcal{L}_{\text{pos}} + \lambda_o \mathcal{L}_{\text{ori}} + \lambda_r \mathcal{L}_{\text{root}} + \lambda_s \mathcal{L}_{\text{smooth}}],$$

where $Q_{1:T}^0$ is the **raw Unitree G1 retarget**. For corresponding body landmarks $\mathcal{J}$,

$$\mathcal{L}_{\text{pos}} = \sum_{j \in \mathcal{J}} w_j \|FK_j(\mathbf{q}_t) - \mathcal{A}(\mathbf{p}_{t,j}^H)\|_2^2,$$

with $\mathcal{A}(\cdot)$ providing embodiment-aware scaling. We emphasize pelvis, feet, and hands because errors at these landmarks most strongly alter stance, support, and movement identity. The final column of each example in Fig. 1 visualizes this raw human-to-G1 transfer.

The raw-retargeting gap. Kinematic fidelity does not imply executable robot motion. **Morphology mismatch** changes reachable stance width and limb extension; **DoF mismatch** limits human hip, spine, shoulder, and ankle configurations; **contact mismatch** induces floating, penetration, or foot slip; and **dynamic mismatch** maps human-valid accelerations and momentum to robot states that may violate support, friction, or actuation limits. These failures are amplified by the movement classes in Table 2, particularly deep stance, unilateral support, rotation, and aerial transition. Thus,

kinematically faithful retargeting $\not\Rightarrow$ physically executable humanoid motion.

The corpus therefore terminates at

Live Video $\rightarrow$ 3D Human Motion $\rightarrow$ Skill/Phase/Contact $\rightarrow$ Raw G1 Retarget Q^0 .

This boundary is deliberate: the corpus provides structured human motion and its raw humanoid embodiment. **KalariSena begins from Q^0** , addressing how to make these references physically grounded, determine whether their intended continuation remains viable, and correct execution before that continuation is lost.

3 KalariSena: Prospective Skill Viability for Humanoid Whole-Body Control

The previous section provides a raw Unitree G1 retarget $Q_{1:T}^0$ together with movement identity, phase, and contact structure. KalariSena asks a prospective question: **can the robot execute the current movement while preserving a physically viable path to its intended continuation?** The method has three stages:

Physics-Grounded Skill $\rightarrow$ Successor-Conditioned Viability $\rightarrow$ Intent-Preserving Correction.

Table 2: **Control taxonomy of the KalariSena corpus.** ✓: dominant characteristic; ✗: not primary. Categories are non-exclusive.

Class	Family	Representative movements	Deep stance	Support Δ	Single support	Angular mom.	Aerial	Long horizon
Stance	Vadivu	Low/guard squat; seated/warrior stance	✓	✗	✗	✗	✗	✗
Footwork	Chuvadu	Square/triangle footwork; stance shifts	✗	✓	✗	✗	✗	✓
Weight transfer	Vadivu/Chuvadu	Warrior lunge; Pakarchakaal transition	✓	✓	✓	✗	✗	✓
Single-leg	Vadivu	Lunge-to-one-leg; unilateral support	✗	✓	✓	✗	✗	✗
Pivot/rotation	Cross-family	Squat pivots; directional transitions	✓	✓	✓	✓	✗	✗
Explosive advance	Chuvadu	Forward attack; deep-squat advance	✓	✓	✗	✓	✗	✗
Jump/landing	Cross-family	Stepping jumps; aerial transitions	✗	✓	✓	✓	✓	✗
Body sequence	Meypayattu	Continuous stance-footwork combinations	✓	✓	✓	✓	✓	✓

The central distinction is

feasible now $\nrightarrow$ viable for the intended continuation.



Figure 2: **Failure under dynamically inconsistent retargeting.** The translucent red body shows the raw motion reference overlaid on Unitree G1. As support becomes infeasible, the robot falls. Bottom plots show raw lowest-foot height and CoM offset; shaded regions mark floating and penetration.

At time t , let

$$\mathbf{s}_t = (\mathbf{q}_t, \dot{\mathbf{q}}_t, \mathbf{x}_t^{\text{base}}, \dot{\mathbf{x}}_t^{\text{base}}, \mathbf{c}_t)$$

denote the robot state, including joint configuration and velocity, floating-base state, and contact state. The structured skill context is

$$\mathbf{z}_t = (k_t, \phi_t, \mathbf{c}_t, \mathbf{d}_t),$$

where k_t is movement identity, ϕ_t is normalized phase, $\mathbf{c}_t$ is the contact mode, and $\mathbf{d}_t$ collects dynamic features such as CoM motion and centroidal momentum. We represent the intended continuation as

$$g_t^+ = (k_t^+, \phi_t^+, \mathbf{c}_t^+),$$

covering either the next phase of the current movement or entry into a subsequent skill. In the representative failure in Figure 2, the raw retarget progressively loses valid ground contact, develops increasing CoM displacement, and ultimately causes the robot to fall.

3.1 Physics-Grounded Skill

Raw retargeting preserves human geometry but may violate robot contact, balance, or actuation constraints, as also observed in recent physics-aware retargeting methods (Jeong et al., 2025; Zhao et al., 2026). We therefore refine Q^0 into a robot-feasible reference

$$Q_{1:T}^* = \mathcal{P}(Q_{1:T}^0, \mathbf{z}_{1:T}; \mathcal{R}),$$

where $\mathcal{R}$ denotes the target humanoid.

We organize the projection objective into fidelity and feasibility:

$$\mathcal{L}_{\text{phys}} = \underbrace{\mathcal{L}_{\text{fidelity}}}_{\text{preserve the skill}} + \lambda_{\text{feas}} \underbrace{\mathcal{L}_{\text{feasibility}}}_{\text{make it executable}},$$

with

$$\mathcal{L}_{\text{fidelity}} = \lambda_p \mathcal{L}_{\text{pose}} + \lambda_e \mathcal{L}_{\text{ee}},$$

and

$$\mathcal{L}_{\text{feasibility}} = \lambda_c \mathcal{L}_{\text{contact}} + \lambda_b \mathcal{L}_{\text{support}} + \lambda_d \mathcal{L}_{\text{dyn}} + \lambda_l \mathcal{L}_{\text{limits}} + \lambda_s \mathcal{L}_{\text{smooth}}.$$

For foot f , contact consistency penalizes tangential slip and non-zero height when support is intended, and ground penetration otherwise:

$$\mathcal{L}_{\text{contact}} = \sum_{t,f} \left[c_{t,f} \left(\|\mathbf{v}_{t,f}^f\|_2^2 + \alpha_h h_{t,f}^2 \right) + (1 - c_{t,f}) \text{ReLU}(-h_{t,f})^2 \right].$$

Pose alone is insufficient to characterize balance during rapid weight transfer. Following capture-point intuition (Pratt et al., 2006), we use

$$\boldsymbol{\xi}_t = \mathbf{p}_{t,\text{com}}^{xy} + \frac{\dot{\mathbf{p}}_{t,\text{com}}^{xy}}{\omega_t}, \quad \omega_t = \sqrt{\frac{g}{h_t}},$$

and penalize

$$d(\boldsymbol{\xi}_t, \mathcal{S}_t)^2,$$

where $\mathcal{S}_t$ denotes the active support region. Joint, velocity, and torque limits, friction constraints, self-collision, and rigid-body consistency complete the feasibility objective. The resulting Q^* is therefore a **physically grounded reference**; KalariSena’s main problem begins when actual execution departs from it.

3.2 Successor-Conditioned Viability

A robot can remain physically stable while already losing the ability to execute its intended next phase. We therefore distinguish two questions: **what constitutes a valid continuation?** and **whether that continuation remains reachable from the current dynamic state.**

What counts as a valid successor? For continuation g , we collect states near its entry from successful training executions:

$$\mathcal{E}_g = \{\eta(\mathbf{s}_i)\}_{i=1}^{N_g},$$

where $\eta(\mathbf{s})$ encodes root-relative pose, joint/base velocity, CoM motion, centroidal momentum, and contact. Features are normalized using statistics estimated from training executions, and $\mathbf{W}$ balances heterogeneous physical quantities.

Rather than requiring arrival at one canonical pose, we define distance to the empirical successor-entry manifold:

$$D_g(\mathbf{s}) = \frac{1}{K} \sum_{\bar{\mathbf{s}} \in \text{KNN}_K(\eta(\mathbf{s}), \mathcal{E}_g)} \|\mathbf{W}(\eta(\mathbf{s}) - \eta(\bar{\mathbf{s}}))\|_2.$$

A state is a valid successor entry when it is dynamically close to successful examples and satisfies the required contact structure:

$$\Omega(g) = \{\mathbf{s} : D_g(\mathbf{s}) \leq \epsilon_g \wedge \Gamma(\mathbf{c}(\mathbf{s}), \mathbf{c}_g) = 1\}.$$

All statistics defining $\Omega(g)$, including feature normalization and successor-entry statistics, are estimated from training data only.

Can the robot still get there? From state $\mathbf{s}_t$, continuation is successful only if the robot remains safe, completes the ongoing movement, and reaches the successor-entry set $\Omega(g_t^+)$:

$$Y_t^{\text{via}} = \mathbf{1} \left[Y_t^{\text{safe}} \wedge Y_t^{\text{cur}} \wedge Y_t^{\text{succ}} \right],$$

where

$$Y_t^{\text{succ}} = \mathbf{1} \left[\exists t' \leq t + H_V : \mathbf{s}_{t'} \in \Omega(g_t^+) \right].$$

Our **Successor-Conditioned Viability Critic (SCVC)** estimates

$$V_\psi(\mathbf{s}_t, \mathbf{z}_t, g_t^+) = P_\pi \left(Y_t^{\text{safe}} = 1, Y_t^{\text{cur}} = 1, Y_t^{\text{succ}} = 1 \mid \mathbf{s}_t, \mathbf{z}_t, g_t^+ \right).$$

SCVC differs from a generic goal-conditioned success predictor in the event it estimates and the structure on which it conditions. Its target requires the conjunction of *physical safety*, completion of the *current structured skill*, and arrival in the entry manifold of the *designated successor*. Moreover, viability is conditioned jointly on skill phase, contact, dynamic state, and successor identity rather than on a terminal goal alone. Consequently, **the same physical state may be viable for one continuation and non-viable for another.** We use “viability” in this empirical, policy-conditioned sense, not as a formal control-theoretic viability kernel.

3.3 Learning the Viability Boundary

Nominal executions contain mostly easy positive states and provide little supervision near the boundary at which continuation is about to become unrecoverable. We therefore construct counterfactual states around nominal trajectories:

$$\tilde{\mathbf{s}}_t = \mathcal{T}(\bar{\mathbf{s}}_t, \delta), \quad \delta \sim p_{\text{pert}}(\delta \mid k_t, \phi_t, \mathbf{c}_t).$$

Perturbations cover external impulses, friction, mass/inertia, actuator strength, latency, state noise, contact timing, and phase. Sampling is balanced across phases and contacts so that SCVC observes difficult regions including support switches, single-support states, high-momentum phases, and landings.

Each perturbed state is rolled forward under the current policy to obtain an empirical binary label Y_t^{via} from the resulting trajectory; SCVC does not generate its own supervision. We optimize

$$\mathcal{L}_V = \underbrace{\mathcal{L}_{\text{BCE}}(V_\psi, Y^{\text{via}})}_{\text{discrimination}} + \lambda_B \underbrace{(V_\psi - Y^{\text{via}})^2}_{\text{Brier calibration}}.$$

The calibration term is important because V_ψ is not used merely for binary success/failure prediction: its magnitude directly controls *when* KalariSena intervenes and *how strongly* corrective control is applied.

3.4 Intent-Preserving Corrective Control

A nominal whole-body policy tracks the physics-grounded reference:

$$\mathbf{a}_t^0 = \pi_0(\mathbf{s}_t, \mathbf{r}_{t:t+H}).$$

When the intended continuation remains viable, the nominal action should remain largely unchanged.

As viability deteriorates, a residual controller proposes

$$\Delta \mathbf{a}_t = \pi_\theta(\mathbf{s}_t, \mathbf{r}_{t:t+H}, \mathbf{z}_t, g_t^+, V_t),$$

and KalariSena executes

$$\mathbf{a}_t = \mathbf{a}_t^0 + g(V_t)\Delta \mathbf{a}_t, \quad g(V) = \text{clip}\left(\frac{\tau_h - V}{\tau_h - \tau_l}, 0, 1\right).$$

Thus $g(V) \approx 0$ for a healthy trajectory and increases toward 1 only as the designated continuation becomes difficult to preserve.

Generic recovery is insufficient because an arbitrary stabilizing step may prevent a fall while destroying the Kalaripayattu movement or making its intended successor unreachable. We therefore quantify deviation from movement intent as

$$D_{\text{skill}} = \alpha_q D_q + \alpha_e D_{ee} + \alpha_v D_v + \alpha_\phi D_\phi + \alpha_c D_c.$$

The residual policy is trained with

$$\begin{aligned} r_t^{\text{KS}} = & \underbrace{w_t r_{\text{track}} + w_c r_{\text{contact}} + w_b r_{\text{balance}}}_{\text{preserve current skill}} \\ & + \underbrace{w_s r_{\text{succ}} + w_v r_{\text{via}}}_{\text{preserve intended future}} \\ & - \underbrace{w_d D_{\text{skill}}}_{\text{penalize intent deviation}} - \underbrace{w_\Delta \|\Delta \mathbf{a}_t\|_2^2}_{\text{minimal correction}}. \end{aligned}$$

The primary prospective reward is actual arrival in $\Omega(g_t^+)$; the frozen critic provides auxiliary shaping rather than defining success itself. KalariSena therefore seeks the **smallest correction that restores a viable continuation**.

3.5 Iterative Learning and Deployment

Corrective control changes the state distribution encountered by the robot, making a critic trained only on nominal trajectories progressively stale. We therefore alternate rollout collection, critic fitting and calibration, and residual-policy improvement:

$$\pi^{(j)} \rightarrow \mathcal{B}^{(j)} \rightarrow V_\psi^{(j+1)} \rightarrow \pi^{(j+1)}.$$

New rollouts expose SCVC to the evolving policy distribution, while replay preserves earlier failure modes. During each policy-update stage, the newly fitted critic is frozen; the following rollout stage then refreshes its training distribution.

At deployment, we temporally filter the critic output to suppress short-lived fluctuations around the intervention thresholds:

$$\bar{V}_t = \beta \bar{V}_{t-1} + (1 - \beta) V_t.$$

We use two thresholds $\tau_l < \tau_h$ to separate nominal tracking, prospective correction, and conservative fallback. Thresholds and filtering parameters are selected using training/validation rollouts and then fixed during evaluation.

The deployed action is

$$\mathbf{a}_t^{\text{deploy}} = \begin{cases} \mathbf{a}_t^0, & \bar{V}_t > \tau_h, \\ \mathbf{a}_t^0 + g(\bar{V}_t)\Delta \mathbf{a}_t, & \tau_l < \bar{V}_t \leq \tau_h, \\ \mathbf{a}_t^{\text{safe}}, & \bar{V}_t \leq \tau_l, \end{cases}$$

where $\mathbf{a}_t^{\text{safe}}$ denotes the conservative fallback controller used when continuation is no longer judged recoverable.

KalariSena therefore never trades safety for forced completion. Its contribution lies in the intermediate regime: **anticipate loss of the intended future and correct the current motion while that future remains recoverable**.

Track $\rightarrow$ Anticipate $\rightarrow$ Correct $\rightarrow$ Fallback.

Implementation details. We use $K = 20$ nearest neighbors, successor horizon $H_V = 50$ control steps (1.0 s), and $\epsilon_g = 0.20$ for successor-entry estimation after feature normalization. SCVC is a 3-layer MLP with hidden dimensions (512, 256, 128) and is trained with AdamW at a learning rate of 1×10^{-4} and batch size 4096. The residual policy uses the same hidden dimensions and is optimized with PPO for 1.2×10^8 simulation steps using 4096 parallel environments. Control runs at 50 Hz. At deployment, we use $\tau_l = 0.25$, $\tau_h = 0.70$, and $\beta = 0.90$; all hyperparameters are selected on validation rollouts and fixed before evaluation.

4 Experiments and Results

We evaluate KalariSena around five questions: **Q1** Does physics grounding improve executability without erasing motion identity? **Q2** Does successor-conditioned viability predict loss of the intended future before tracking failure? **Q3** Does prospective correction preserve skill intent under disturbance? **Q4** Does it generalize to held-out transitions between known skills and to out-of-distribution perturbations? **Q5** Do these gains transfer to physical humanoid hardware?

4.1 Experimental Setup

Evaluation splits. We use three complementary test splits from the live-recorded corpus. **Motion-held-out** contains unseen executions of movement

classes observed during training. **Transition-held-out** retains both endpoint skills k_i and k_j during training but withholds the directed connection $k_i \rightarrow k_j$; it therefore tests compositional generalization of *how known skills connect*, rather than generalization to an unseen skill. Finally, **perturbation-OOD** extends disturbance magnitude, direction, timing, or robot dynamics beyond their training ranges.

Control perturbations. We perturb execution through

$$\mathcal{D}_{\text{pert}} = \{\mathbf{J}_{\text{ext}}, \mu, m, \mathbf{I}, s_{\tau}, \Delta t, \epsilon_s, \Delta \phi\},$$

covering external impulse, friction, mass/inertia, actuator strength, latency, observation noise, and phase error. Disturbances are injected at dynamically sensitive phases: stance loading, support switch, single support, peak momentum, and landing.

Matched evaluation protocol. All methods use the same humanoid embodiment, reference motions, observations, control rate, domain randomization, perturbation conditions, and evaluation seeds. Projection methods use the same frozen tracker; viability critics use matched capacity and training rollouts; robustness controllers receive the same perturbation at the same phase of the same reference. We report absolute differences in percentage points unless explicitly identified as relative reductions.

4.2 Baselines

We compare methods targeting the major stages of the problem. For **motion retargeting**, we use Raw Retargeting, our physics-grounded projection, CORE (Jeong et al., 2025), and NMR (Zhao et al., 2026). For **whole-body tracking**, we include compatible implementations of KUNGFUBOT (Xie et al., 2025), KUNGFUBOT2 (Han et al., 2025), SONIC (Luo et al., 2026), and BEYONDMIMIC (Liao et al., 2026). For **robustness and recovery**, we compare with ANY2TRACK (Zhang et al., 2026), KUNGFUATHLETEBOT (Lei et al., 2026), FIRM (Xu et al., 2026), STABLEMIMIC (Wu et al., 2026), **Track+Fallback**, and **Track+Recovery**. For **skill transitions**, we additionally include SWITCH (Lau et al., 2026). We report only methods reproducible or adaptable to the same humanoid embodiment under the common protocol above.

Baseline fairness. All baselines use the same Unitree G1 embodiment, motion references, observations, control rate, domain randomization, perturbation instances, and test seeds. Projection methods share the same frozen tracker, while viability critics use matched capacity and training rollouts. Results requiring unmatched embodiments or evaluation settings are excluded.

4.3 Metrics

Fidelity and physical feasibility. We report mean per-joint position error (MPJPE), root/end-effector error, velocity error, foot slip, contact accuracy, joint/torque violations, self-collision, penetration, and fall rate. Movement-intent deviation is

$$E_{\text{skill}} = \frac{1}{T} \sum_t D_{\text{skill}}(\mathbf{s}_t, \mathbf{r}_t).$$

Continuation and intent preservation. Completion requires safe completion of the current skill; transition success additionally requires reaching the valid successor-entry set $\Omega(g^+)$. Their conjunction defines **Intent Preservation Rate (IPR)**:

$$\text{IPR} = \frac{1}{N} \sum_{i=1}^N \mathbf{1} \left[Y_{\text{safe}}^{(i)} \wedge Y_{\text{cur}}^{(i)} \wedge Y_{\text{succ}}^{(i)} \right].$$

Thus, merely preventing a fall or returning the robot to a neutral state does not count as intent preservation.

Prospective viability. For the Successor-Conditioned Viability Critic (SCVC), we report AUROC, AUPRC, Brier score, expected calibration error (ECE), and anticipation lead time

$$T_{\text{lead}} = t_{\text{fail}} - t_{\text{detect}},$$

measuring how early loss of the intended continuation is detected relative to observed tracking failure.

4.4 Q1: Does Physics Grounding Improve Executability?

We isolate reference quality from controller quality by evaluating all retargeting methods with the same frozen whole-body tracker.

Table 3(a) shows that raw kinematic fidelity alone is insufficient for executable humanoid motion. Physics grounding reduces constraint violations from 18.4% to **5.7%** (**69.0% relative reduction**) and foot slip from 8.7 to **2.9 cm/s** (**66.7% relative reduction**).

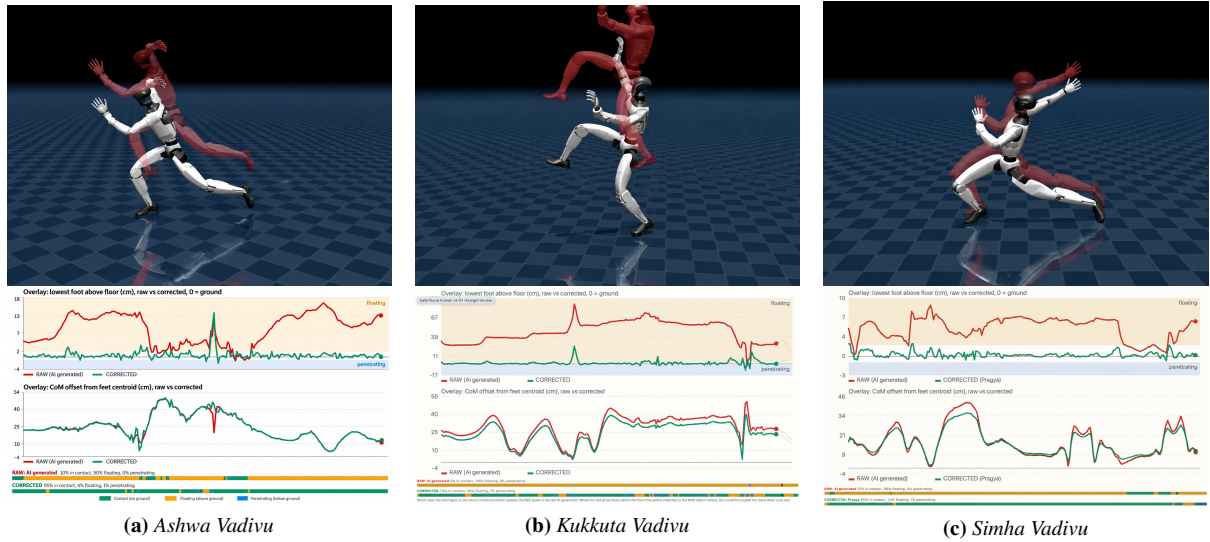


Figure 3: **Qualitative KalariSena results on representative Kalaripayattu movements.** For each example, the top image overlays the human reference (translucent red) and Unitree G1 realization (white). The bottom plot reports lowest-foot height above ground and CoM offset from the feet centroid for raw and corrected motions. Across all three movement regimes, physics-grounded correction improves contact consistency and dynamic support while retaining characteristic movement structure.

Table 3: **Component validation of KalariSena.** (a) Physics-grounded projection on held-out motions. (b) Prospective viability prediction with matched critic capacity and training rollouts; lead time is in ms.

(a) Physics-grounded projection					
Method	Slip↓ (cm/s)	Viol.↓ (%)	MPJPE↓ (mm)	Comp.↑ (%)	$E_{\text{skill}} \downarrow$
Raw Retargeting	8.7	18.4	42.1	61.3	0.118
CoRe	5.2	11.6	45.8	72.4	0.126
NMR	4.8	10.1	43.7	75.9	0.121
Ours: Physics	2.9	5.7	43.4	84.6	0.109

(b) Prospective viability prediction					
Critic	AUROC↑	AUPRC↑	Brier↓	ECE↓	Lead↑
State	0.761	0.708	0.194	0.082	94
+ Phase/contact	0.823	0.776	0.166	0.061	126
+ Future reference	0.867	0.831	0.143	0.047	158
SCVC	0.921	0.892	0.108	0.026	214

This physical consistency costs little geometric fidelity: MPJPE changes from 42.1 mm to 43.4 mm (**3.1%**), while E_{skill} improves from 0.118 to **0.109**. Closed-loop completion rises from 61.3% to **84.6%** (**+23.3 points**), showing that the gain comes from removing embodiment-induced physical inconsistencies rather than smoothing away movement identity.

4.5 Q2: Does SCVC Predict the Intended Future?

The key question is narrower than whether a neural critic can predict future success: does *explicit knowledge of the designated successor*

add predictive information beyond physical state, phase/contact structure, and a generic future-reference window? KalariSena uses viability in this empirical, policy-conditioned sense, not as a formal control-theoretic viability kernel.

We compare capacity-matched critics trained on identical rollouts:

$$\begin{aligned}
 V_{\text{state}} &= V(s_t), \\
 V_{\text{phase}} &= V(s_t, \phi_t, c_t), \\
 V_{\text{ref}} &= V(s_t, \mathbf{r}_{t:t+H}), \\
 V_{\text{SCVC}} &= V(s_t, \mathbf{z}_t, g_t^+).
 \end{aligned}$$

Table 3(b) shows that SCVC reaches **0.921 AUROC**, versus 0.867 for the strongest successor-agnostic critic (**+0.054**), while AUPRC rises from 0.831 to **0.892**. Calibration also improves: Brier score falls from 0.143 to **0.108**, and ECE from 0.047 to **0.026**.

SCVC detects impending loss of continuation **56 ms earlier**, increasing anticipation lead time from 158 ms to **214 ms**. This matters around support switches, single-support states, and high-momentum phases, where pose error can remain small even after the intended successor is becoming dynamically unreachable.

Why successor conditioning matters. The critic comparison isolates its predictive contribution, while the closed-loop successor ablation in Table 5(a) tests its effect on control. Removing

successor conditioning reduces held-out transition success from 83.7% to 70.6% (**−13.1 points**), perturbed-transition success from 76.4% to 63.1% (**−13.3 points**), and IPR from 74.8% to 61.9% (**−12.9 points**), while seen-transition performance changes by only 1.7 points. This asymmetry supports the central claim: viability depends on *which future the robot intends to preserve*.

4.6 Q3: Does Prospective Correction Preserve Intent?

We next test whether anticipation improves closed-loop behavior. Each controller receives the same perturbation at the same phase of the same motion reference.

Table 4 separates *surviving a disturbance* from *preserving the intended movement future*. Recovery-oriented baselines reduce falls, but a stabilizing step or return to neutral can make the designated continuation unreachable.

KalariSena achieves **81.6% IPR**, versus 66.8% for KungFuAthleteBot (**+14.8 points**). Transition success rises from 69.5% to **84.1% (+14.6 points)**, while current-skill completion reaches **90.8%**.

These gains do not sacrifice motion fidelity: MPJPE is 44.9 mm, only 1.1 mm above the base tracker. KalariSena simultaneously achieves the lowest fall rate (**6.7%**), shortest recovery time (**0.46s**), and fewest torque violations (**3.4%**).

Relative to the base tracker, IPR improves by **36.9 points** (44.7%→81.6%), transition success by **34.8 points** (49.3%→84.1%), and completion by **24.7 points** (66.1%→90.8%), while falls drop by **22.9 points**. The dominant gain is therefore not merely stabilization, but stabilization that preserves access to the intended successor.

4.7 Q4: Generalization Beyond Training

Held-out skill connections. Both endpoint skills k_i and k_j are observed during training, while the directed transition $k_i \rightarrow k_j$ is withheld; this tests unseen connections between known skills, not unseen-skill generalization.

As shown in Table 5(a), KalariSena reaches **83.7%** held-out transition success, versus 74.2% for SWITCH (**+9.5 points**). The difference is smaller on seen transitions (93.2% versus 90.1%), indicating that prospective successor reasoning is most valuable when the transition itself was not observed during training.

Controlled successor ablation. Table 5(a) also provides a capacity-controlled comparison with **Ours w/o successor**. Removing successor conditioning changes seen-transition success by only **−1.7 points** (93.2%→91.5%) but reduces held-out success by **13.1 points** (83.7%→70.6%) and IPR by **12.9 points** (74.8%→61.9%). Thus, the gain is not explained simply by additional model capacity or better nominal tracking.

Perturbation-OOD. Under perturbed transitions, Table 5(a) shows that KalariSena reaches **76.4%** success, outperforming SWITCH (61.4%) by **15.0 points** and the successor-agnostic ablation (63.1%) by **13.3 points**. IPR reaches **74.8%**, versus 59.8% for SWITCH.

As disturbance severity increases, control exhibits three regimes: at low severity, competent trackers remain near the nominal trajectory; at intermediate severity, prospective correction becomes increasingly active; at extreme severity, viability falls below τ_l and control transfers to the conservative safety policy. The operating principle is

preserve intent when recoverable; fallback when it is not.

For operating threshold τ , we characterize this trade-off using

$$\text{Coverage}(\tau) = P[V_\psi > \tau], \quad \text{Risk}(\tau) = P[\text{failure} \mid V_\psi > \tau].$$

A useful viability estimate should lower risk at matched coverage, rather than merely assign high confidence more frequently.

4.8 Ablation Study

Table 5(b) decomposes the contribution of physical grounding, structured skill state, prospective viability, explicit successor conditioning, and viability-gated residual correction.

Physics grounding raises IPR from 44.7% to 52.8% (**+8.1 points**); phase/contact skill structure raises it to 59.1% (**+6.3**); prospective viability to 67.5% (**+8.4**); explicit successor conditioning to 74.6% (**+7.1**); and viability-gated residual correction to **81.6% (+7.0)**.

Overall, KalariSena improves IPR by **36.9 percentage points** over the base tracker. The progression is diagnostic: physics grounding makes the reference executable; structured skill state identifies where the robot is in the movement; viability provides a prospective failure signal; successor

Table 4: **Robustness and intent preservation under matched perturbations.** IPR requires physical safety, current-skill completion, and entry into the designated successor.

Method	MPJPE↓ (mm)	Fall↓ (%)	Comp.↑ (%)	Trans.↑ (%)	IPR↑ (%)	Rec. Time↓ (s)	Torque Viol.↓ (%)
Base Tracker	43.8	29.6	66.1	49.3	44.7	0.91	11.8
Any2Track	45.1	18.7	76.5	59.8	55.6	0.73	8.6
Track+Fallback	48.6	10.9	78.2	52.4	48.9	0.58	5.2
Track+Recovery	49.3	9.8	82.6	63.7	60.8	0.64	5.9
KungFuAthleteBot	46.7	11.4	83.9	69.5	66.8	0.61	6.3
KalariSena	44.9	6.7	90.8	84.1	81.6	0.46	3.4

Table 5: **Generalization and ablation of KalariSena.**

(a) Generalization to held-out skill connections and perturbations. (b) Incremental component ablation. Success and IPR values are in %; P/C denotes phase/contact.

(a) Held-out transition generalization				
Method	Seen↑	Held-out↑	Pert.↑	IPR↑
Base Tracker	82.4	61.7	48.9	46.3
BeyondMimic	87.6	68.5	56.8	53.7
Switch	90.1	74.2	61.4	59.8
Ours w/o succ.	91.5	70.6	63.1	61.9
KalariSena	93.2	83.7	76.4	74.8

(b) Component ablation					
Variant	Phys.	P/C	Succ.	Via.	Res. IPR↑
Base Tracker	—	—	—	—	44.7
+ Physics	✓	—	—	—	52.8
+ Skill structure	✓	✓	—	—	59.1
+ Viability	✓	✓	—	✓	67.5
+ Successor	✓	✓	✓	✓	74.6
KalariSena	✓	✓	✓	✓	81.6

Table 6: **Physical-robot results across five movement/transition regimes under nominal and perturbed conditions.** All values are percentages.

Method	Nominal↑	Perturbed↑	IPR↑	Falls↓
Base Tracker	88.3	59.6	47.9	24.2
Track+Recovery	86.7	72.1	61.5	10.8
KalariSena	91.4	83.2	77.6	5.4

conditioning asks whether the *designated* continuation remains reachable; and the residual policy converts that signal into an intent-preserving correction. No individual component accounts for the full gain.

4.9 Q5: Sim-to-Real Deployment

We deploy the same policy architecture on the physical humanoid across **5 movement/transition regimes**: deep stance, weight transfer, single support, rotation, and dynamic transitions, under **2 matched conditions**: nominal and perturbed. All methods use identical motion references and pre-defined safe perturbations.

Table 6 shows that KalariSena reaches **91.4%**

nominal success, versus 88.3% for the base tracker and 86.7% for Track+Recovery. Under disturbance, success improves from 72.1% to **83.2%** (**+11.1 points**) and IPR from 61.5% to **77.6%** (**+16.1 points**) over Track+Recovery, while falls decrease from 10.8% to **5.4%** (**50% relative reduction**).

Relative to the base tracker, KalariSena improves perturbed execution by **23.6 points** and IPR by **29.7 points**. The larger gains under perturbation are consistent with the viability gate: corrections remain small on healthy trajectories and increase near the viability boundary. Typical corrections include support-foot adjustment, torso counter-rotation, and swing retiming; beyond the recoverable region, SCVC activates conservative fallback.

4.10 Qualitative Analysis

Figure 3 illustrates the transition from raw human-to-humanoid retargeting to physics-grounded execution. Across Ashwa Vadivu, Kukkuta Vadivu, and Simha Vadivu, corrected motions maintain more consistent ground contact and reduce excessive CoM displacement while preserving the defining stance and whole-body structure.

These observations agree with Table 3(a): constraint violations and foot slip fall substantially, while the change in MPJPE is small and E_{skill} decreases. The correction therefore does not simply produce a more conservative or upright movement.

The robustness results in Table 4 expose the second distinction: generic recovery can preserve physical stability without preserving the skill. An additional support step may prevent a fall yet produce a state from which the designated next stance or transition is no longer reachable. KalariSena instead restores dynamic margin while keeping the robot within the set of states capable of completing the current movement and entering its intended successor.

The central experimental distinction is therefore

survive the disturbance $\neq$ *preserve the intended skill future*.

Across projection quality, prospective prediction, matched perturbations, held-out transitions, component ablations, and physical-hardware deployment, the results consistently support the latter objective.

5 Conclusion

We introduced **KalariSena**, a physics-grounded framework for transferring highly dynamic **Kalari-payattu** skills to humanoid robots, built from approximately **100 hours of live-recorded demonstrations**. KalariSena combines (i) **structured skill representation** over movement, phase, contact, and dynamics, (ii) **physics-grounded embodiment projection** enforcing support, contact, friction, and actuation constraints, and (iii) **Successor-Conditioned Viability with viability-gated residual control** to anticipate loss of the intended continuation and apply minimal, intent-preserving corrections before fallback.

The results support the central premise that **feasibility now does not imply viability for what comes next**. Under matched perturbations, KalariSena improves completion from **66.1% to 90.8%**, reduces physical violations by **69.0%**, and raises intent preservation from **44.7% to 81.6%**. It further reaches **83.7%** success on held-out transitions and **77.6%** IPR on physical hardware. Together, these results suggest a broader view of humanoid robustness: **robust control should preserve not only the present motion, but a viable path to its intended future**. This perspective extends naturally to other agile, structured whole-body skills.

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